Low Pressure Chemical Vapor Deposition (LPCVD)

Brian Chu

Department of Nanoengineering
University of California,
San Diego

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[uwaterloo.ca]

UC San Diego
JACOBS SCHOOL OF ENGINEERING

This presentation focuses on exploring which parameters have a statistically significant effect on the deposition rate

[lamresearch.com]

Concept and Applications

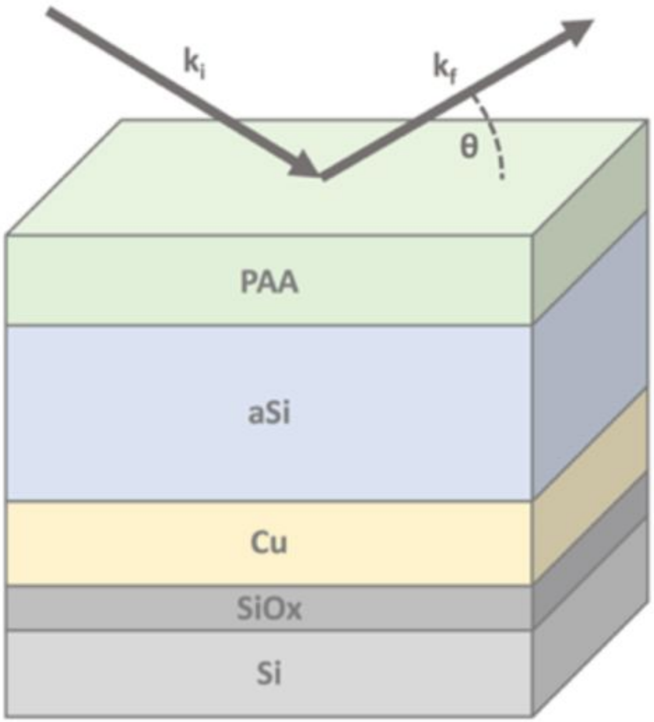
[comsol.com]

COMSOL
Experiment Design

[medium.com]

Results and Statistical Analysis

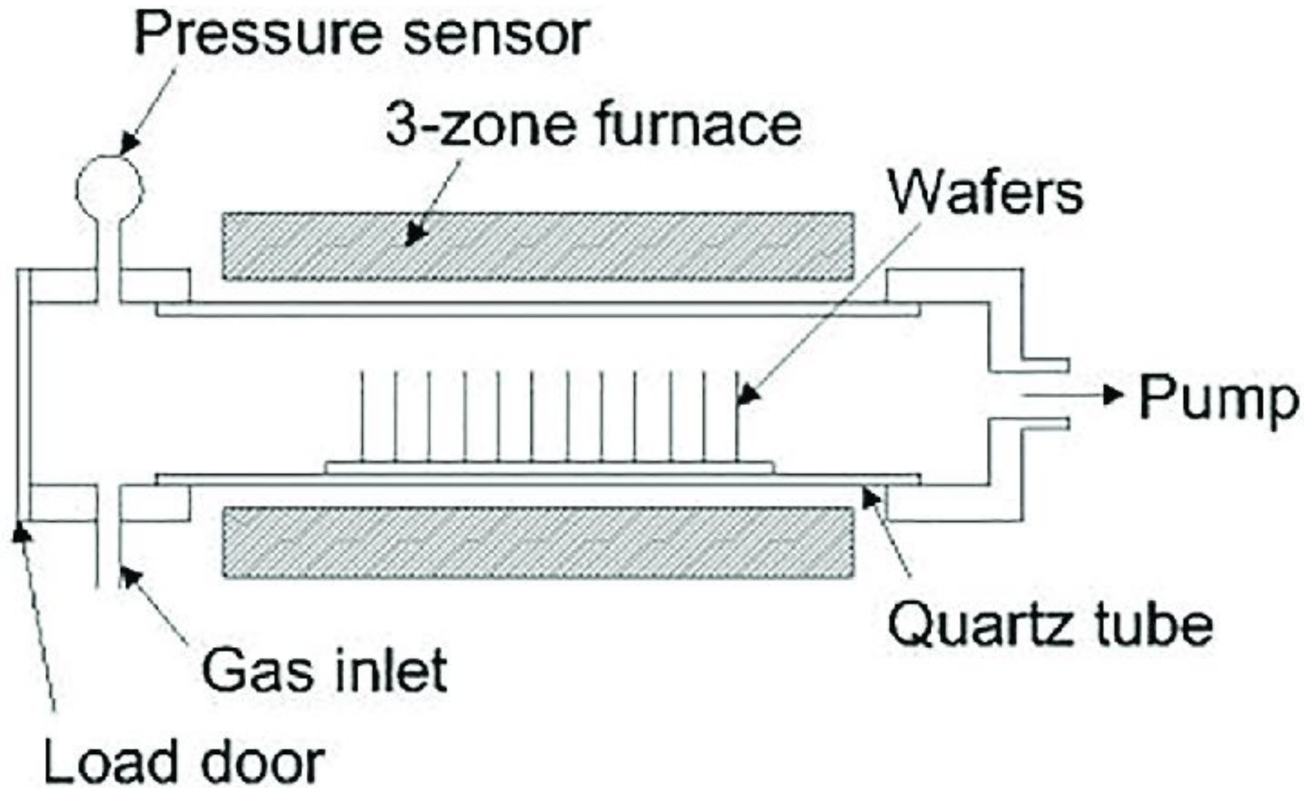
Semiconductor manufacturing use chemical vapor deposition to create precise thin film layers



[Doucet, 2017]

- polysilicon
- silicon nitride
- silicon dioxide
- silicon oxynitride

LPCVD uses surface reactions and diffusion in forming protective insulation



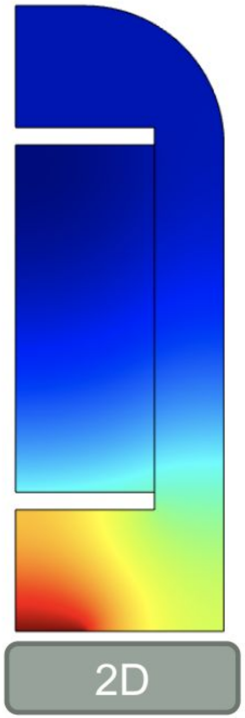
[Hajare, 2021]

$$SiH_4 + O_2 \rightarrow SiO_2 + 2H_2$$

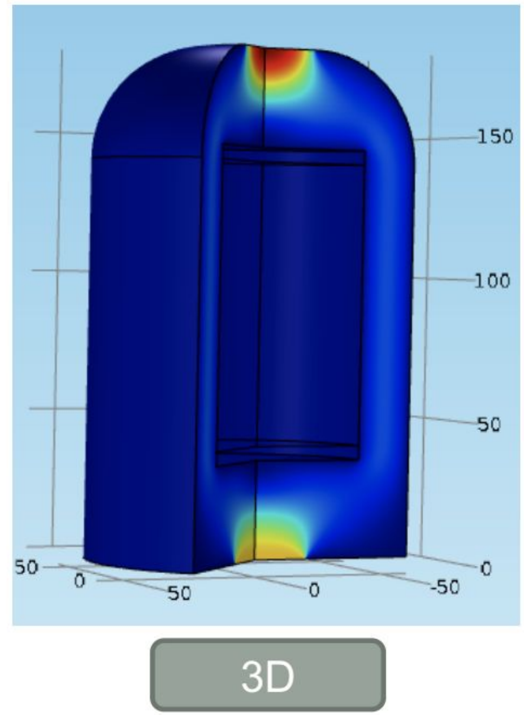
Why Chemical Vapor Deposition at Low Pressure?

- **uniform film**
- **high purity**
- **reproducibility**
- **homogeneity**

Simulations done on COMSOL Multiphysics



2D Model of a
Boat Reactor



3D Model of a
Boat Reactor

[Dr Yang Introduction to COMSOL]

In the Design of Experiment (DoE), 5 parameters were varied to study their effects on deposition

A → Molar fraction of SiH_4
B → Temperature of Section 1
C → Distance between wafers
D → Inlet Velocity
E → Total pressure

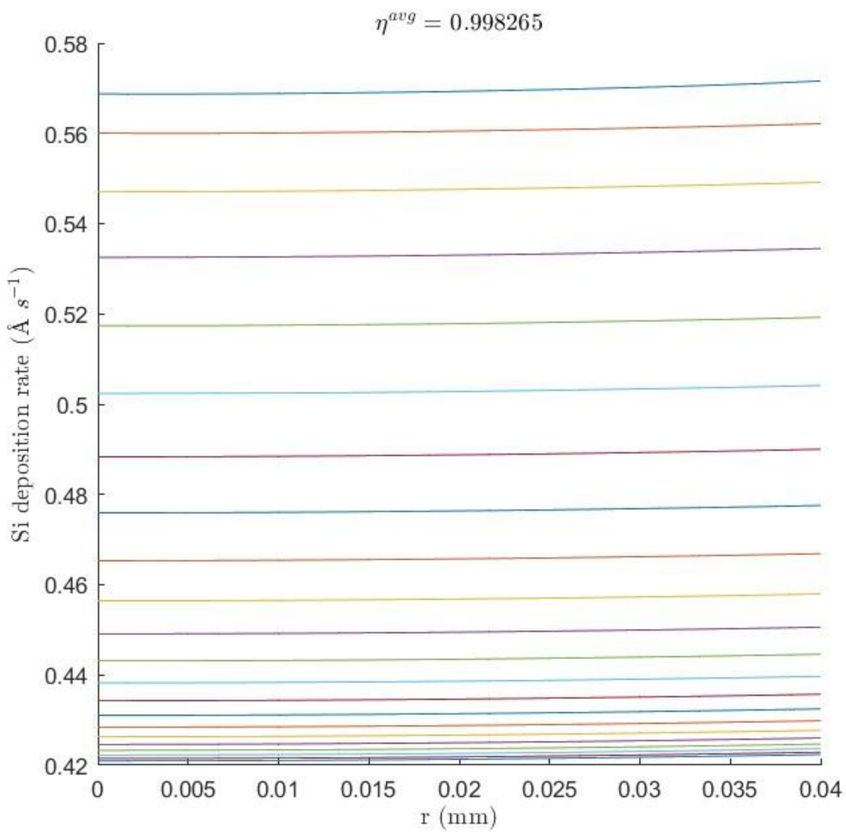
→ COMSOL runs combinations

$$Effectiveness\ factor = \frac{Average\ deposition\ rate\ of\ wafer}{Edge\ deposition\ rate}$$

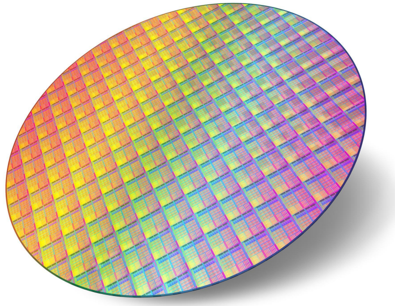
COMSOL

[comsol.com]

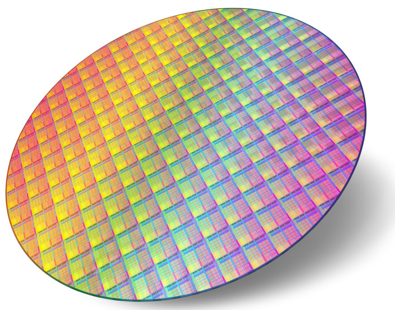
Curve shows deposition rate for each simulation



[Experimental Data]



First wafer gas contacts



Last wafer gas contacts

[waferworld.com]

The significance of each parameter was evaluated

	Deposition Rate	Effectiveness Factor
Variable	**P Value**	**P Value**
A (Mole Fraction)	1.876E-31	9.303E-23
B (Temperature)	1.115E-52	3.891E-34
C (Wafer Spacing)	2.219E-22	3.347E-22
D (Inlet Velocity)	.3821	.8475
E (Total Pressure)	1.411E-28	1.118E-9

Variable	P Value	P Value
AB	1.659E-25	1.696E-17
AC	9.625E-6	1.254E-6
AD	.9640	.9825
AE	2.248E-4	.9519
BC	5.063E-19	1.849E-17
BD	.5952	.9092
BE	3.901E-21	5.5769E-3
CD	.9633	.9793
CE	.7542	.7280
DE	.9660	.9904

Statistical analysis shows that only 4 of the 5 factors alone had significant effects on average effectiveness

8 of the 15 combinations yielded significant change in average effectiveness factors

The factors alone of

Mole Fraction
Temperature
Wafer Spacing
Total Pressure

The combinations of

Mole fraction + Temperature
Mole fraction + Wafer Spacing
Temperature+ Wafer Spacing
Temperature + Total Pressure

In summary, 8 of the 15 combinations are proven to have a significant effect on the average effectiveness

Reaction rate and free silicon atoms are major factors for increasing deposition rate

⬇

Temperature and Mole Fraction have the most significant effects

[angstromengineering.com]

Questions?